

DRAFT SAMPLE EXAMINATION MARKING KEY

DRAFT

INTEGRATED SCIENCE STAGE 2—ANALYTIC MARKING KEY

SECTION ONE

Question Number	Answer
1.	a
2.	a
3.	d
4.	b
5.	b
6.	c
7.	d
8.	c
9.	c
10.	a
11.	d
12.	d
13.	b
14.	a
15.	b
16.	b
17.	a
18.	c
19.	c
20.	c

SECTION TWO

Question 1

(a)

Description	Marks
sensory neuron	1
connector neuron,	1
motor neuron	1
effector	1
Total	4

(b)

(i)

Description	Marks
stimulation of a sensory receptor causes	1
electrochemical impulse to move along membrane	1
Total	2

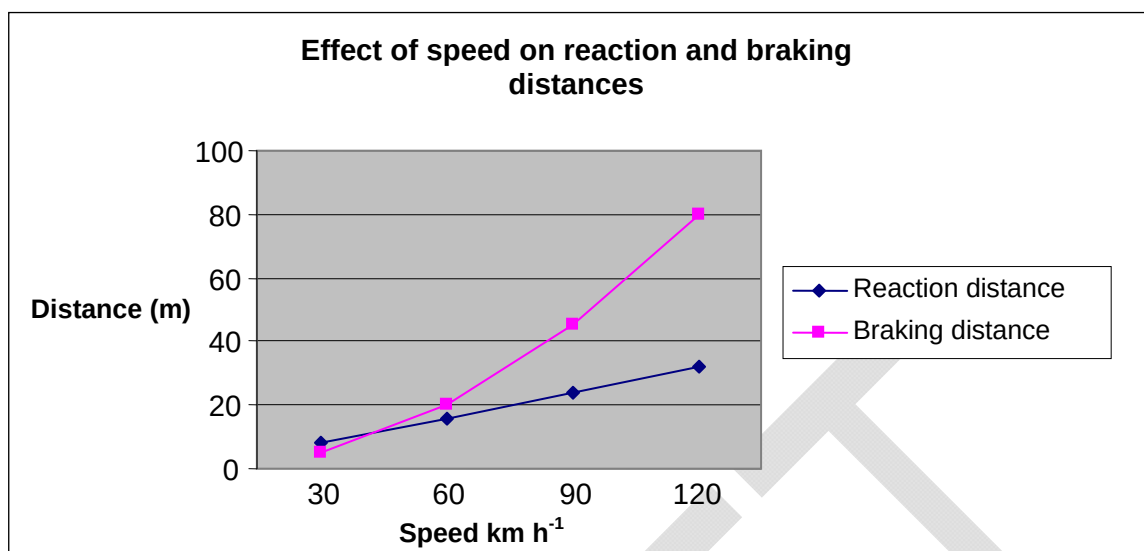
(ii)

Description	Marks
chemical transmitter/neurotransmitter molecules released	1
bridge gap and bind to specific receptors	1
Total	2

(c)

Description	Marks
Any three of the following: • anticipation • experience • fatigue (tiredness) • distractions (music, passengers, mobile phone) • visibility • drugs (alcohol, medication, illicit drugs) • age • eyesight • hearing	1 mark each
Total	3

(d) Need numbers on X axis



Description	Marks
correct axes—labelled	1
scale	1
reaction distance	1
braking distance	1
Total	4

(e)

Description	Marks
reaction time increases by about 2.5 m for every 10 km h ⁻¹	1
stopping distance increases disproportionately/at a more rapid rate	1
Total	2

(f)

Description	Marks
Any two of the following:	
• speed warning signs	1
• visible police presence on road as deterrent	1
• use of speed cameras	
Total	2

Question 2

(a)

Description	Marks
The ice cream container without the lid i.e. test 1	1
This records the sound level as the ring tone has travelled through the air to the microphone with no interference to its transmission.	1
Total	2

(b)

Description	Marks
He used a musical ring tone, which is not a constant frequency	1
He should record and use a single note ring tone	1
Total	2

(c)

Description	Marks
Any four valid modifications (1 mark each): • Multiple trials with each material • Use more materials • Use different amounts of each material • Ensure that all variables apart from material are kept constant (e.g. volume of phone, microphone distance etc) • Use more sophisticated equipment (e.g. audio oscillator as sound source etc) • Any consideration that looks at reflection, or absorption or transmission separately • use same thickness of materials	1 mark each
Total	4

(d)

Description	Marks
All other tests had a 10dB result range. Neoprene only had a 7dB range	1
Neoprene must have absorbed different frequencies in the ring tone more effectively	1
Total	2

(e)

Description	Marks
Have ring tones that are at a number of specific frequencies (e.g. 250Hz, 500Hz, 1000Hz, 2000Hz)	1
Use a stand to hold the microphone at a constant distance from the phone, and repeat the procedure described in part (c) for each of the frequencies.	1
Record all data collected in an appropriate format (e.g. table, or graph)	1
Total	3

(f)

Description	Marks
The gymnasium has many hard reflective surfaces	1
Reflected sound is uninterrupted on its path to the ear	1
If the original and reflected sounds are far enough apart (1/10 s) then an echo is heard	1
Student's bodies are soft and absorb the original and reflected sounds	1
Total	4

(g)

Description	Marks
Concert hall A has a regular reflection that focuses all sound to one point	1
Concert hall B has diffuse reflection that 'spreads' the sound evenly through the auditorium	1
Concert hall B is best	1
Total	3

Question 3

(a)

Description	Marks
• sound focused/collected by auricle • transmitted through air in external auditory canal • tympanic membrane vibrates • making the malleus move • ossicles act as levers to transmit movement across middle ear • stapes vibrates against oval window in cochlea • movement in fluid in cochlea • hair cells are stimulated • nerve impulses transmitted	any 6 connected points
Total	6

(b)

Description	Marks
8 cm/4 waves	1
2 cm	1
Total	2

(c)

Description	Marks
amplitude: produces more energy—loudness increases	1
frequency: higher frequency may go beyond upper limit of human hearing range	1
Total	2

(d)

Description	Marks
$v = f \lambda$	1
$346 = 2000\lambda$	
$\lambda = 346/2000$	1
$\lambda = 0.173$	
$\lambda = 17.3 \text{ cm}$	1
Total	3

(e)

	Conductive deafness	Nerve deafness
Cause in relation to how the ear works	Any one of the following: • blockage of the external auditory canal prevents sound reaching the tympanic membrane • infection causing accumulation of fluid in the middle ear not allowing full conduction of sound • fusing of the ossicles so that vibration cannot be transmitted	defects or damage of the sensory hairs in the cochlea or auditory nerve
Treatment	Appropriate treatment for cause given: • removal of blockage • treatment of infection • hearing aids transmit sound through the bone of the skull	cochlear implant/no treatment

Description	Marks
Conductive deafness—1 cause	1
Conductive deafness—1 treatment (appropriate to cause listed)	1
Nerve deafness—1 cause	1
Nerve deafness—1 treatment	1
Total	4

(f)

Description	Marks
industrial noise—particular frequency	1
damages hair cells in one region of the cochlea	1
nerve impulse not transmitted for this region/frequency	1
Total	3

Question 4

(a)

Description	Marks
screening—grit removal	2
sedimentation—settling undissolved substances	2
Total	4

(b)

Description	Marks
aerobic digestion:	
• bacterial digestion in presence of oxygen	1
• aeration tanks/air injected by blowers on floor of tank	1
anaerobic digestion:	
• bacterial action in the absence of oxygen	1
• produces methane and carbon dioxide	1
Total	4

(c)

Description	Marks
insoluble iron phosphate not available for plant use	1
does not add to nutrient load of ocean where it is discharged	1
Total	2

(d)

Description	Marks
sand filtering—does not remove all toxic materials	2
chlorine—may not kill all pathogenic organisms	2
Total	4

(e)

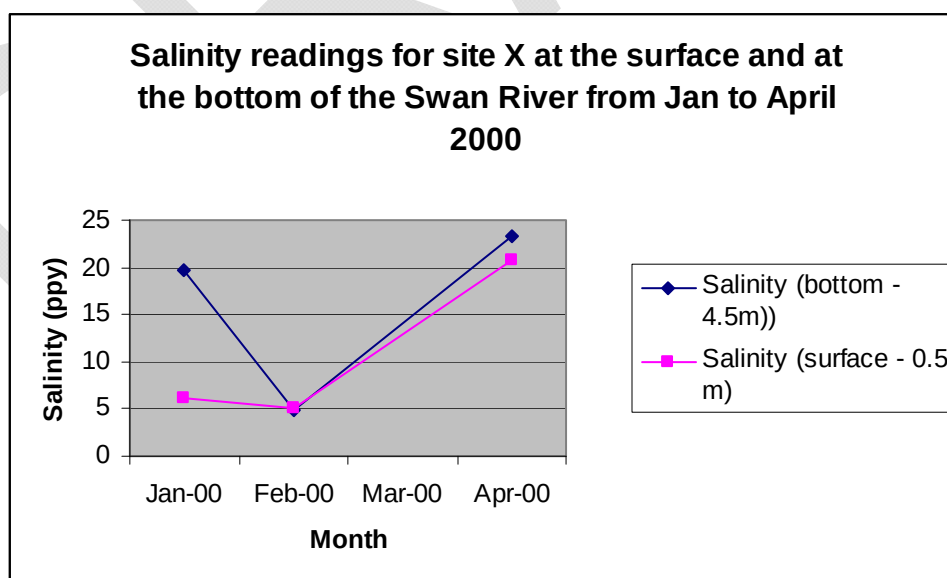
Description	Marks
parks and gardens	1
recharge aquifers	1
Total	2

(f)

Description	Marks
reverse osmosis process uses membranes with very fine pores	1
water is a very small molecule and can be forced through the membrane	1
larger molecules e.g. pathogens, hormones	1
cannot pass through the membrane	1
Total	4

Question 5

(a)



Description	Marks
correctly labelled axes	1
correctly scaled axes	1
points plotted accurately	1
depth identified using key/label	1
Total	4

(b)

Description	Marks
salt water moves upstream as a wedge	1
it sits below the freshwater due to higher density	1
Total	2

(c)

(i)

Description	Marks
large rain events cause freshwater to enter the system	1
dilution and mixing causes salinity to fall at the surface and at depth	1
Total	2

(ii)

Description	Marks
water flushed out to sea	1
salt wedge has moved up the river past this site	1
tidal exchange causes mixing of water/reduces stratification	1
Total	3

(d)

Description	Marks
important food source for other organisms	1
converts light energy to biomass/base of food pyramid/supports food web	1
Total	2

(e)

Description	Marks
lowered salinity favoured <i>Microcystis</i>	1
nutrient levels increased due to run-off from farms	1
Total	2

(f)

Description	Marks
long day length—availability of light for growth	1
warm temperatures/clear calm waters	1
Total	2

(g)

Description	Marks
determine extent of bloom and plan control/management measures	1
provide information to public regarding current status and health implications e.g. warnings not to swim in particular areas	1
Total	2

(h)

Description	Marks
Any one of the following: • increase the salinity of the surface layer of the water beyond the tolerance of the algae • mechanical removal of algae using a skimmer • using flocculating material to try and sink the algae	1
Total	1

Question 6

(a)

Description	Marks
oxygen—animals respiration	1
temperature—lower oxygen levels/stress	1
conductivity—salinity osmotic gradient	1
Total	3

(b)

Description	Marks
recognise that 130 million litres is 17% of total	1
calculate 100%	1
divide by number of people (1.6 million)	1
correct answer = 478 litres per person per day	1
Total	4

(c)

Description	Marks
labelled diagram (semi-permeable membrane, compartments with concentrated and dilute solutions)	2
high pressure is applied to the compartment containing sea water	1
large amount of energy needed for pump which supplies pressure	1
compartments separated by a membrane with a very fine pore structure	1
Total	5

(d) (i)

Description	Marks
definition—process used to remove salt from salt water	1
Total	1

(ii)

Description	Marks
less rainfall—unable to fill dams	1
groundwater limited supply	1
desalination is non weather dependent	1
Total	3

(e)

Description	Marks
Any two: map plant and animal species identify endangered species and unique species identify areas for seed collection inform of appropriate revegetation	1 mark each
Total	2

(f)

Description	Marks
Any two: plants would be suited to the local conditions/maintain soil stability encourage native fauna back into the area native animals would assist in pollination of plants aesthetic/heritage value	1 mark each
Total	4

SECTION THREE

(a)

Description	Marks
For each point listed, one mark allocated. For each corresponding coherent explanation one mark allocated.	max 10 marks

Students may include any of the following points:

Description
In relation to the following points, discuss the effect on the reaction time and distance and how this relates to the stopping distance • drink driving • drug use • fatigue • distractions e.g. mobile phones, billboards, passengers In relation to the following points, discuss the effect on braking distance and how this relates to stopping distance. • speeding • road conditions e.g. wet/slippy roads, gravel roads • vehicle maintenance e.g. tyre condition, brakes

(b)

Description	Marks
For each point listed, one mark allocated. For each corresponding coherent explanation one mark allocated.	maximum 10 marks

Description
• describe Newton's First Law of Motion (inertia) • when the car stops, an unrestrained occupant continues to move forward • define momentum • unrestrained occupant has greater momentum when hitting windscreen/dashboard • seat belt slows down the forward movement due to the elasticity of the seat belt/less momentum • seatbelt distributes the resultant force due to large surface area • crumple zones take time to crumple • slow the car and reduce the momentum • not stopping instantly • occupant in the car also has less momentum

Question 8

Students may include any of the following points. Each point should be well explained for a maximum of 2 marks per point.

Description	Marks
• most significant impact on reducing water consumption has been the introduction of the watering days regime • using more water efficient plants in the garden • different types of watering e.g. drippers rather than sprinklers • education campaigns through advertising on TV and brochures included with water accounts • Waterwise Rebate Program—cash back for water efficiency including wetting agents for gardens, approved front loader washing machines, water efficient shower heads, • Waterwise Rebate Program –cash back for rainwater tanks to use water that would otherwise not be available • reusing grey water for gardens using strict guidelines	maximum 8 marks
• underground water systems to reduce evaporation • genetically modified plants to reduce need for water but maintain productivity • changing to crops that do not require large amounts of water when it is not available	maximum 6 marks
• Australia dry continent/low freshwater supplies • total annual rainfall decline 10% over past 30 years • manage sources of water • increase diversity of supply—dams, groundwater, desalination, recycling • use of recycled water for purposes other than drinking so that drinking water supplies can be maintained • conservation measures implemented to reduce need to increase supply	maximum 6 marks
Total	20

Integrated Science: Sample Examination Stage 2
Mapping questions to content

Quest. No	The practice of science		Conceptual understandings						The impact of science	
	2A	2B	2A Adolescents and driving	2A Adolescents and music	2B Importance of water	2B Properties of water	2B Aquatic ecosystems	2B Water resources and sustainability	2A	2B
Section One—Multiple Choice										
1						✓				
2		✓								
3								✓		
4						✓				
5						✓				
6							✓			
7							✓			
8						✓				
9						✓				
10							✓			
11									✓	
12			✓							
13			✓							
14			✓							
15			✓							
16			✓							
17	✓									
18				✓						
19				✓						
20				✓						

Quest. No	The practice of science		Conceptual understandings						The impact of science	
	2A	2B	2A Adolescents and driving	2A Adolescents and music	2B Importance of water	2B Properties of water	2B Aquatic ecosystems	2B Water resources and sustainability	2A	2B
Section Two—Short Answer										
1			✓						✓	
2	✓			✓					✓	
3				✓					✓	
4								✓		✓
5		✓					✓			✓
6						✓		✓		✓

Quest. No	The practice of science		Conceptual understandings						The impact of science	
	2A	2B	2A Adolescents and driving	2A Adolescents and music	2B Importance of water	2B Properties of water	2B Aquatic ecosystems	2B Water resources and sustainability	2A	2B
Section Three—Extended Answer										
1			✓						✓	
2								✓		✓

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